ALL MATHS FORMULA SHEET

Quick Reference Guide for Competitive Exams & Revision

1 ALGEBRA

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$
- $(a + b)(a - b) = a^2 - b^2$
- $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
- $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

2 QUADRATIC EQUATIONS

Standard Form

$$ax^2 + bx + c = 0 \quad (a \neq 0)$$

Quadratic Formula

$$\boxed{x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}}$$

Discriminant

$D = b^2 - 4ac$

- $D > 0$ → two distinct real roots
- $D = 0$ → two equal real roots
- $D < 0$ → two complex roots

3 TRIGONOMETRY

Basic Ratios

$\sin\theta = P / H$	$\operatorname{cosec}\theta = H / P$
$\cos\theta = B / H$	$\sec\theta = H / B$
$\tan\theta = P / B$	$\cot\theta = B / P$

Important Identities

- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \operatorname{cosec}^2\theta$
- $\tan\theta = \sin\theta / \cos\theta$
- $\cot\theta = \cos\theta / \sin\theta$
- $\sec\theta = 1 / \cos\theta$
- $\operatorname{cosec}\theta = 1 / \sin\theta$

4 GEOMETRY

Triangle

- $Area = {}^1/_2 \times b \times h$
- $Perimeter = AB + BC + CA$
- $s = {}^{(a + b + c)}/_2$
- $Area\ (Heron's) = \sqrt{[s(s-a)(s-b)(s-c)]}$

Parallelogram

- $Area = b \times h$
- $Perimeter = 2(a + b)$

Circle

- $Circumference = 2\pi r$
- $Area = \pi r^2$
- $Arc\ Length = (\theta / 360°) \times 2\pi r$
- $Sector\ Area = (\theta / 360°) \times \pi r^2$

5 COORDINATE GEOMETRY

Distance Formula

$PQ = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

Section Formula

P divides $Q_1(x_1,y_1)$, $Q_2(x_2,y_2)$ in ratio m:n

$P = \left({}^{(mx_2+nx_1)}/_{(m+n)}\ ,\ {}^{(my_2+ny_1)}/_{(m+n)} \right)$

Midpoint Formula

$M = \left({}^{(x_1+x_2)}/_2\ ,\ {}^{(y_1+y_2)}/_2 \right)$

Slope

$m = (y_2 - y_1) / (x_2 - x_1)$

Equation of a Line

- Point-Slope: $y - y_1 = m(x - x_1)$
- Slope-Intercept: $y = mx + c$
- Two-Point: ${}^{(y - y_1)}/_{(y_2 - y_1)} = {}^{(x - x_1)}/_{(x_2 - x_1)}$

6 MENSURATION (3D)

Cube (side = a)

- $Volume = a^3$
- $TSA = 6a^2 \mid LSA = 4a^2$

Cuboid (l, b, h)

- $Volume = lbh$
- $TSA = 2(lb + bh + hl)$
- $LSA = 2h(l + b)$

Cylinder (r, h)

- $CSA = 2\pi rh \mid TSA = 2\pi r(h + r)$
- $Volume = \pi r^2 h$

Cone (r, h, slant height = l)

- $CSA = \pi rl \mid TSA = \pi r(l + r)$
- $Volume = {}^1/_3\, \pi r^2 h$

Sphere (r)

- $Surface\ Area = 4\pi r^2$
- $Volume = {}^4/_3\, \pi r^3$

7 MENSURATION (2D)

Square (side = a)

- $Area = a^2$
- $Perimeter = 4a$

Rectangle (l, b)

- $Area = l \times b$
- $Perimeter = 2(l + b)$

Parallelogram (b, h)

- $Area = b \times h$
- $Perimeter = 2(a + b)$

Rhombus (diagonals d_1, d_2)

- $Area = {}^1/_2\, d_1 d_2$
- $Perimeter = 4a$

Trapezium (a, b parallel; h)

- $Area = {}^1/_2\, (a + b)h$
- $Perimeter = a + b + c + d$

8 STATISTICS & PROBABILITY

Statistics

- Mean $(\bar{x}) = {}^{\Sigma x}/_n$
- **Median:** Middle value (arranged)
- **Mode:** Most frequent value
- **Range:** Highest – Lowest

Probability

$$\boxed{P(E) = \frac{\text{No. of favorable outcomes}}{\text{Total no. of outcomes}}}$$

$$0 \leq P(E) \leq 1$$

Special Cases

- $P(E) = 0$ → Impossible event
- $P(E) = 1$ → Certain event
- $P(E) = {}^1/_2$ → Equally likely event

9 LOGARITHMS

- $\log(ab) = \log a + \log b$
- $\log({}^a/_b) = \log a - \log b$
- $\log(a^n) = n \log a$
- $\log_a b = {}^{\log b}/_{\log a}$

10 SEQUENCES & SERIES

Arithmetic Progression (A.P.)

- $a_n = a + (n - 1)d$
- $S_n = {}^n/_2\, [2a + (n - 1)d]$

Geometric Progression (G.P.)

- $a_n = ar^{n-1}$
- $S_n = {}^{a(r^n - 1)}/_{(r - 1)} \quad (r \neq 1)$

11 DIFFERENTIATION

${}^d/_{dx}\,(x^n) = n\,x^{n-1}$	${}^d/_{dx}\,(\tan x) = \sec^2 x$
${}^d/_{dx}\,(\sin x) = \cos x$	${}^d/_{dx}\,(e^x) = e^x$
${}^d/_{dx}\,(\cos x) = -\sin x$	${}^d/_{dx}\,(\log x) = {}^1/_x$

12 INTEGRATION

$\int x^n\,dx = {}^{x^{n+1}}/_{(n+1)} + C \quad (n \neq -1)$

$\int {}^1/_x\,dx = \log|x| + C$

$\int \cos x\,dx = \sin x + C \qquad (C = \text{Constant of Integration})$

$\int e^x\,dx = e^x + C$

$\int \sin x\,dx = -\cos x + C$